ICMP: The Internet Control Message Protocol

1 Overview

The **Internet Control Message Protocol (ICMP)**, specified in RFC 792, is used by hosts and routers to communicate network-layer information. Its primary use is **error reporting** (e.g., "Destination network unreachable").

Architecture

- **Layering:** Architecturally sits just above IP. ICMP messages are carried as **IP payload**.
- **Protocol Number:** IP datagrams with ICMP specify an upper-layer protocol number of **1**.
- **Structure:** messages have a **Type** and a **Code** field. They also contain the header and first 8 bytes of the datagram that triggered the error.

2 ICMP Message Types

Type	Code	Description
0	0	Echo reply (to ping)
3	0	Destination network unreachable
3	1	Destination host unreachable
3	3	Destination port unreachable
4	0	Source quench (congestion control - rare)
8	0	Echo request
11	0	TTL expired
12	0	IP header bad

Table 1: Selected ICMP Message Types.

3 Practical Applications

Ping

- Sends an ICMP **Type 8, Code 0** (Echo Request).
- Destination responds with **Type 0, Code 0** (Echo Reply).
- Most operating systems support the ping server directly in the kernel (not as a separate process).

Traceroute

Traceroute is implemented using ICMP and UDP probes:

1. Source sends UDP segments with increasing **TTL** (1, 2, 3...).
2. Each router along the path decrements TTL. When it hits 0, the router discards the packet and sends an ICMP **Type 11, Code 0** (TTL expired) back to the source.
3. The source learns the router's identity from the ICMP message header.
4. **Stopping Criteria:** One packet eventually reaches the destination. Since it uses an unlikely UDP port, the host sends an ICMP **Type 3, Code 3** (Port unreachable), telling the source to stop.

4 ICMPv6

Specified in RFC 4443, ICMPv6 reorganizes legacy types and adds new ones required for IPv6 functionality:

- **Packet Too Big:** Crucial for path MTU discovery in IPv6.
- **Unrecognized IPv6 Options:** New error codes for header issues.